



Fragmentation typologies, trajectories, and path dependence in urban mangrove landscapes of Southeast Asia

Allen Glen Cumaya Gil · Karen C. Seto

Received: 23 January 2026 / Accepted: 16 July 2026
© The Author(s) 2026

Abstract

Context In rapidly urbanizing coastal regions, mangrove ecosystems are increasingly fragmented by cumulative human and natural disturbances. Fragmentation is widely recognized as a dynamic process shaped by historical legacies. However, it is still most often assessed using snapshot-based measurements that obscure the spatiotemporal pathways through which fragmentation develops. As a result, it remains difficult to distinguish among transformation pathways and to determine how prior fragmentation states influence subsequent patch outcomes.

Objectives This study aims to (1) characterize urban mangrove fragmentation as a set of distinct spatiotemporal change signatures using patch-level typologies; (2) identify dominant fragmentation pathways and their association with patch outcomes across diverse urban landscapes; and (3) examines how the history of fragmentation influences subsequent patch outcomes by quantifying whether earlier states provide information beyond the immediately preceding state.

Methods Mangrove fragmentation was analyzed across 21 Southeast Asian urban areas from 1996 to 2020 using patch-level landscape metrics derived from Global Mangrove Watch data. Fragmentation

typologies were identified using a Random Forest classification approach and linked across time using a maximum-overlap tracking framework to reconstruct multi-step patch trajectories. Path dependence was evaluated within a closed cohort of baseline-traceable patch trajectories using an information-theoretic framework. Entropy reduction (ΔH) was used to measure how much earlier fragmentation states reduced uncertainty about later patch outcomes beyond the information provided by the immediately preceding state.

Results Across urban areas, mangrove patches consistently became smaller, more edge-exposed, geometrically simpler, and more isolated, indicating widespread intensification of fragmentation. However, similar trends in aggregate metrics arose from different spatiotemporal transformation pathways. These differences were captured by four dominant fragmentation typologies—Nibbling, Clearing, Shattering, and Displacing—which varied across urban areas in both frequency and spatial extent. Fragmentation trajectories were highly uneven, with a small number of recurring pathways accounting for most observed sequences among baseline-traceable patch trajectories. Pathways dominated by Shattering and Clearing were consistently associated with higher probabilities of patch loss, indicating that specific sequences of spatiotemporal transformations are linked to degradation outcomes. Entropy-based results showed that earlier fragmentation states provided measurable, though limited, additional

A. G. C. Gil (✉) · K. C. Seto
Yale School of the Environment, Yale University,
New Haven, CT, USA
e-mail: allen.gil@yale.edu

information about later patch outcomes ($\Delta H \approx 0.11$ – 0.16 bits), indicating weak-to-moderate but stable historical constraint.

Conclusions Findings from urban mangrove systems in Southeast Asia show that fragmentation unfolds through recurring sequences of spatiotemporal transformations that shape patch outcomes over time. Identifying fragmentation pathways provides a clearer basis for understanding landscape change and associated risks than snapshot metrics alone. Path dependence quantifies how much earlier states inform later outcomes, offering complementary insight into fragmentation dynamics. Together, the typology- and trajectory-based framework provides a transferable approach for diagnosing fragmentation pathways and identifying high-risk pathways in urban and other human-modified landscapes.

Keywords Landscape fragmentation · Mangrove forests · Urbanization · Path dependence · Spatiotemporal trajectories · Southeast Asia

Introduction

Mangrove forests in rapidly urbanizing coastal regions function as critical natural infrastructure. They provide ecosystem services such as shoreline stabilization, storm protection, and carbon sequestration (Bimrah et al. 2022). These services are especially valuable in dense urban settings, where mangroves protect large populations, built infrastructure, and livelihoods (Reyes et al. 2024). Climate change and sea-level rise further increase their importance, particularly for low-lying and resource-constrained communities with limited adaptive capacity (Maggan et al. 2022). Southeast Asia contains the largest share of the world's mangroves and is also a hotspot of rapid land-use change driven by aquaculture expansion, agricultural conversion, and urban growth (Richards and Friess 2016; Goldberg et al. 2020). These pressures, together with natural disturbances such as tropical cyclones, coastal erosion, and sedimentation, have driven sustained mangrove loss and increasing fragmentation across the region.

Beyond areal loss, mangrove fragmentation—the subdivision of continuous forests into smaller, more isolated patches—is an important dimension of ecosystem degradation. In this study, a mangrove patch is

defined as a spatially contiguous group of mangrove pixels and serves as the fundamental unit for analyzing fragmentation dynamics. Fragmentation is related to, but distinct from, mangrove loss. Landscapes may retain stable or even increasing total mangrove area while becoming more fragmented (Bryan-Brown et al. 2020). These structural changes can reduce the effectiveness of key ecosystem services, as smaller and more dispersed patches provide lower wave attenuation, weaker erosion control, and reduced capacity for carbon storage and habitat provision (Henderson et al. 2019; Best et al. 2022; Rani et al. 2023). Fragmentation therefore represents a critical consideration for mangrove conservation and management aimed at sustaining ecosystem service provision.

Despite its importance, fragmentation is most often assessed using broad-scale landscape metrics applied to discrete temporal snapshots, with limited attention to historical dynamics (McGarigal et al. 2016). These approaches summarize landscape condition but do not capture how fragmentation unfolds through time or how prior states influence subsequent patch outcomes (Wan 2025). Similar trends in commonly used metrics can arise from different spatiotemporal transformations. For example, two patches may exhibit comparable declines in area and increases in isolation. One may result from gradual edge retreat, while the other reflects internal subdivision into smaller fragments. These pathways differ in their underlying mechanisms and ecological consequences, even when summary metrics appear similar (McGarigal and Cushman 2002; Cushman et al. 2008; Fahrig 2013; Haddad et al. 2015). Because metric-based assessments cannot distinguish among these pathways, understanding fragmentation dynamics requires linking spatial change with its temporal sequence. This also requires evaluating whether prior fragmentation states provide measurable information about subsequent patch outcomes.

Landscape ecology has long emphasized that fragmentation reflects spatial processes shaped by disturbance, recovery, and reorganization through time (McGarigal and Cushman 2002; McGarigal et al. 2005; Turner 2010). Historical disturbances can leave persistent legacy effects on patch structure and spatial configuration, influencing landscape dynamics across multiple temporal scales (Peters et al. 2011; Turner and Gardner 2015). Despite these theoretical advances, empirical applications have rarely

translated this process-based perspective into practice. Few studies explicitly track temporal sequences or quantify the influence of prior states on fragmentation dynamics. In addition, many commonly used metrics are strongly correlated with habitat amount, limiting their ability to distinguish among different fragmentation pathways when composition and configuration are not explicitly separated (Rutledge 2003; Wang et al. 2014; Hadley and Betts 2016; Adler and Jedicke 2022). These limitations are particularly evident in mangrove systems, where large-scale assessments have focused on areal change and broad fragmentation patterns, with limited attention to the sequencing of patch-level transformations or the pathways shaping fragmentation trajectories (Bryan-Brown et al. 2020).

A related but less explored dimension of fragmentation dynamics is historical contingency, often described as path dependence. Path dependence refers to the influence of past states on future outcomes, as accumulated structural, ecological, or institutional conditions shape system trajectories (Zariņa, 2013). This concept has been widely studied in land-use change and urban systems, where long development histories influence present-day spatial patterns (Roy Chowdhury and Turner 2019; Carlucci and Salvati 2023). In fragmentation research, however, path dependence is rarely quantified explicitly, particularly at the patch level (McGarigal et al. 2005; Zhang et al. 2024). Although prior states are often assumed to influence subsequent outcomes, the extent to which fragmentation history reduces uncertainty about patch outcomes remains unclear. When this historical influence is not evaluated, fragmentation outcomes are implicitly treated as independent of prior states, an assumption that is seldom tested. Quantifying this dependence is therefore important. If fragmentation processes are historically structured, patch outcomes cannot be inferred from current configuration alone. Instead, prior states provide measurable information about subsequent outcomes, which can be quantified to assess the extent of historical influence on fragmentation dynamics (Turner 2010; Johnstone et al. 2016).

Urban mangrove systems are especially likely to exhibit historically structured fragmentation due to the persistence of built infrastructure, cumulative land-use change, and long-term alterations to coastal geomorphology and hydrology (Branoff

2020; Goldberg et al. 2020; Wu et al. 2024). Roads, ports, aquaculture ponds, and flood-control structures create durable spatial constraints that influence how mangrove patches fragment and reorganize over time (Harris et al. 2010; Erfteimeijer et al. 2021). These constraints can shape fragmentation pathways by limiting how landscapes respond to disturbance and recovery processes. In such settings, early fragmentation processes may influence later outcomes, including the likelihood of continued degradation or patch loss. However, comparative studies that jointly examine fragmentation pathways and historical contingency across multiple urban mangrove systems remain limited (Bryan-Brown et al. 2020). As a result, it remains unclear whether observed fragmentation patterns reflect short-term responses or the cumulative effects of historically structured trajectories (Tappeiner et al. 2021).

In response to these gaps, this study examines urban mangrove fragmentation across Southeast Asia as a historically structured and spatially explicit phenomenon. Specifically, this study addresses three related questions: (1) how fragmentation can be characterized as discrete spatiotemporal change signatures at the patch level; (2) which fragmentation pathways dominate across urban mangrove systems; and (3) how fragmentation history influences subsequent patch outcomes and how this varies across urban contexts. To address these questions, we develop a typology of fragmentation signatures using patch-level landscape metrics, temporal change variables, lineage tracking, and machine learning classification. We then reconstruct multi-step, baseline-traceable patch trajectories over time to identify recurrent fragmentation pathways and their associations with patch outcomes. Finally, we apply an information-theoretic framework to quantify historical dependence as a reduction in uncertainty about patch outcomes along these trajectories. This approach separates the frequency of fragmentation pathways from the strength of historical constraint. Together, these analyses provide a process-oriented perspective on urban mangrove fragmentation that links spatial patterns, temporal trajectories, and historical constraints across diverse urban contexts.

Methods

Analytical overview

The analytical workflow consisted of ten sequential steps (Fig. 1). First, mangrove extent and urban boundary data were assembled. Second, mangrove extent data were preprocessed through filtering, raster-to-vector conversion, minimum patch-size screening, and patch delineation. Third, patch-level landscape metrics were calculated for each time step. Fourth, patch lineages were reconstructed across time using a maximum-overlap tracking framework that preserved split events and simplified merge events. Fifth, lineage-derived metrics were calculated to quantify changes in patch structure and position. Sixth, fragmentation typologies were classified using a Random Forest model, and model performance was evaluated through internal validation and external validation with independent urban areas. Seventh, typology assignments were linked over time to construct multi-step fragmentation trajectories that represent sequences of spatiotemporal transformations for each patch. Eighth, inter-urban fragmentation patterns and typology composition were summarized using both proportional and absolute measures. Ninth, path dependence was evaluated by estimating transition probabilities and entropy-based metrics for a closed cohort of baseline-traceable patch trajectories. Tenth,

pathway-specific probabilities of patch loss were estimated from fragmentation trajectories, and uncertainty and sensitivity analyses were conducted using urban-area-blocked bootstrap resampling and alternative sample specifications.

Study area

This study examines mangrove fragmentation across 21 coastal urban areas in Southeast Asia, spanning Indonesia, Malaysia, the Philippines, Thailand, and Vietnam, and including the transboundary Singapore–Johor Bahru urban region (Fig. 2). These urban areas represent a wide range of coastal geomorphic settings, mangrove extents, and levels of urban development, providing a diverse basis for comparison.

In this study, ‘urban mangroves’ refers operationally to mangrove patches located within or near morphologically defined urban areas, including adjacent peri-urban coastal zones. The term identifies mangrove patches associated with urban coastal landscapes, where built-up expansion, infrastructure, land conversion, hydrological modification, and other urban or peri-urban pressures may influence fragmentation dynamics. This definition provides a consistent basis for comparing mangrove fragmentation across countries with different administrative definitions of urban areas.

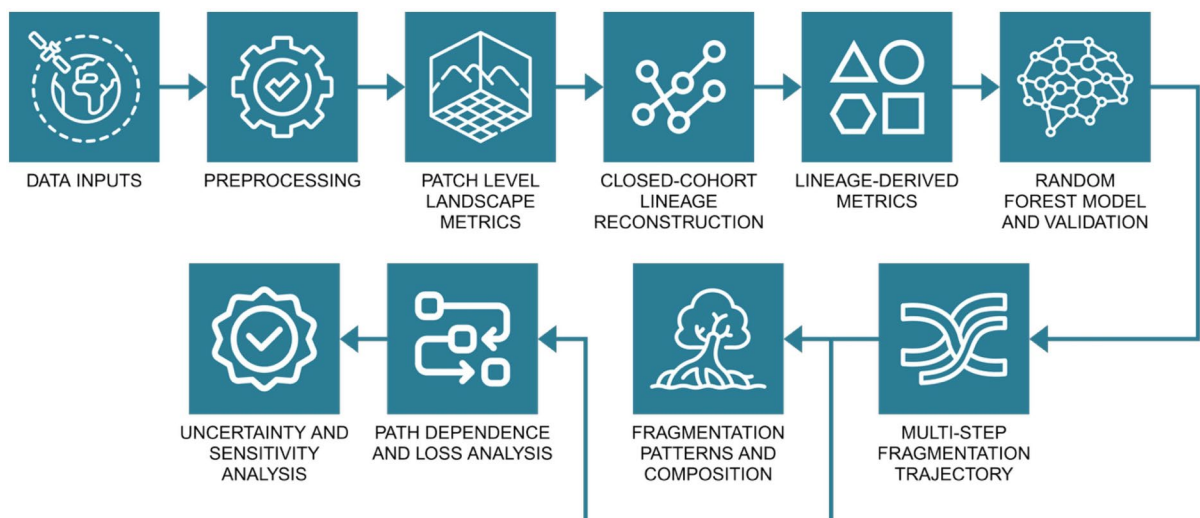


Fig. 1 Analytical workflow for mangrove fragmentation typology classification, trajectory reconstruction, and path dependence analysis

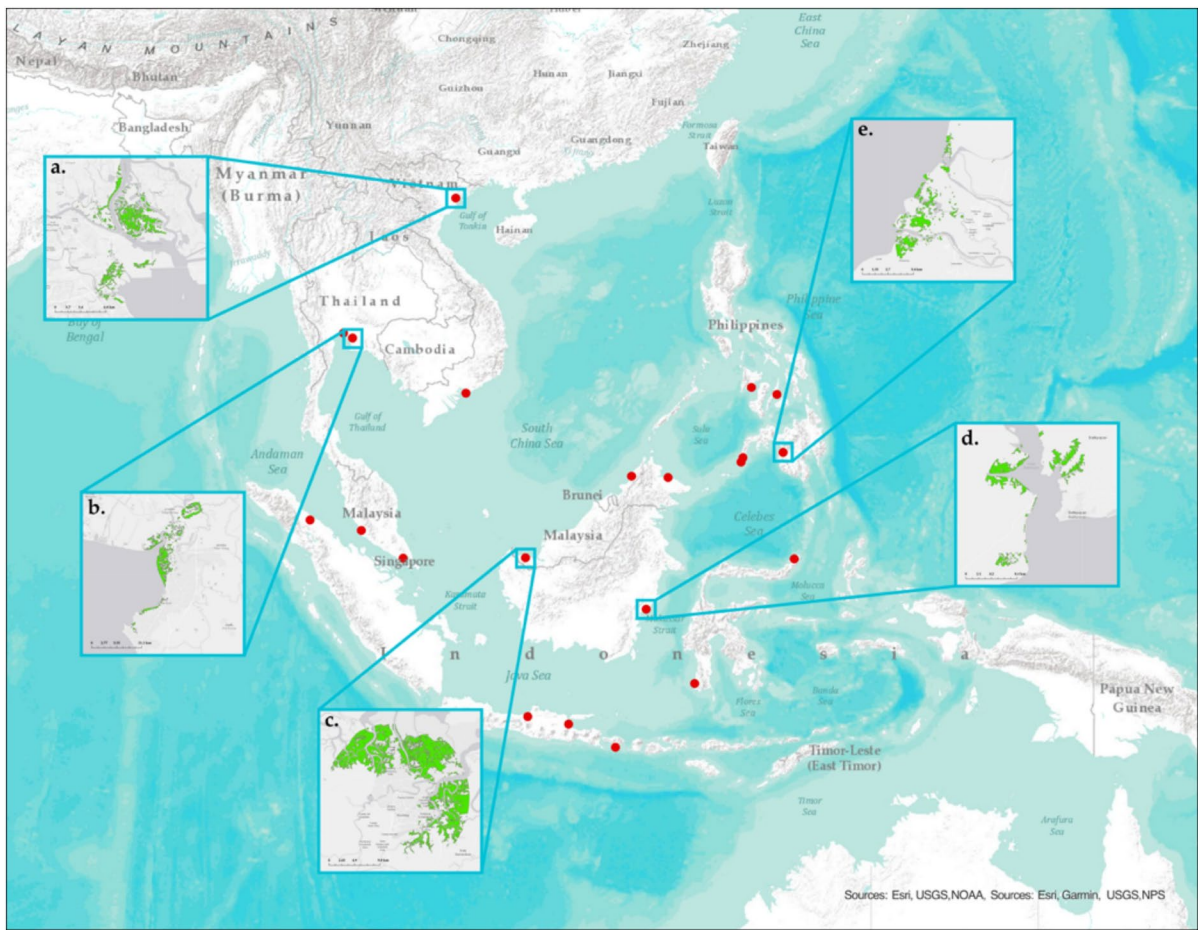


Fig. 2 Map of the 21 study urban areas in Southeast Asia with selected snapshots. **a** Haiphong (Vietnam); **b** Chon Buri (Thailand); **c** Kuching (Malaysia); **d** Balikpapan (Indonesia); **e** Cotabato (Philippines)

The analysis is conducted at the urban-area scale, with each urban area treated as a distinct landscape unit. This design enables comparison across heterogeneous coastal and urban contexts. Within each urban area, fragmentation is examined at the patch level, allowing detailed analysis of how individual mangrove patches are transformed, subdivided, displaced, or lost over time. This hierarchical structure—patch-level analysis nested within urban-area units—provides a consistent framework for comparing fragmentation patterns, trajectories, and historical dependence across urban areas.

Because nearby patches are spatially correlated, they are not treated as independent observations. Instead, statistical inference is conducted at the urban-area level. Urban areas are treated as independent

units due to their geographic separation and lack of spatial continuity.

Mangrove extent and urban boundary data

Mangrove extent data were obtained from the Global Mangrove Watch (GMW) v3.0 dataset (Bunting et al. 2022), provided as raster data with 25-m pixel spacing, for four analytical time points: 1996, 2007, 2015, and 2020. Although GMW is provided as raster data, the analytical unit in this study is the mangrove patch, defined as a spatially contiguous group of mangrove pixels.

GMW v3.0 was selected because it provides a consistent global mangrove extent and change time series across 11 epochs from 1996 to 2020, from which the

four analytical time points used in this study were drawn. The product was developed using L-band Synthetic Aperture Radar (SAR) global mosaics from the Japan Aerospace Exploration Agency and a map-to-image change-detection approach, in which the GMW v2.5 baseline map was updated using thresholding and a contextual mangrove change mask (Bunting et al. 2022). This common product framework supports temporal comparison across years. However, mapping uncertainty remains, especially for gain and loss classes and fragmented settings affected by SAR mosaic misregistration or confusion with adjacent land uses. Observed patch changes are therefore interpreted as product-consistent estimates within the spatial and temporal limits of GMW v3.0.

To support patch-level analysis, each raster was first processed with a majority filter to reduce classification noise, then converted to vector polygons. Each polygon represents an individual mangrove patch. The majority filter was implemented in ArcGIS Pro using a 3×3 neighborhood with a four-neighbor contiguity rule. Pixels were reassigned only when the dominant class among orthogonally adjacent neighbors exceeded the threshold. This approach reduces isolated noise while preserving patch boundaries and minimizing distortion of narrow features.

This preprocessing step involves a trade-off. The majority filter improves patch delineation and temporal tracking by reducing isolated pixels that may reflect classification noise. It may also smooth or omit very small, narrow, or isolated mangrove features, including true patches with ecological value. We therefore interpret the results as estimates of fragmentation dynamics among consistently mapped and traceable patches within GMW v3.0. The analysis does not provide a complete inventory of all small mangrove features within each urban area.

These preprocessing steps established a consistent spatial grain across time periods and urban areas. Vectorization enabled calculation of patch geometry and spatial relationships using ArcGIS Pro. Each patch was assigned a unique identifier to support metric calculation and temporal tracking. Patches smaller than 0.1 ha were excluded to improve the reliability of patch delineation and temporal tracking. This threshold is close to the minimum mapping unit implied by the 25 m GMW pixel spacing, so it mainly removes one-pixel features and very small vector slivers. Such features are especially sensitive

to classification error, georegistration uncertainty, raster-to-vector conversion, and single-pixel changes. They can also disproportionately affect patch-count, edge, and nearest-neighbor metrics. The threshold supports more comparable estimates of fragmentation dynamics across years.

Subsequent analyses focus on relative change, typology transitions, and lineage-conditioned trajectories. Because preprocessing was applied consistently across all time periods and urban areas, its effects are expected to be systematic and comparable across the dataset.

Urban areas were delineated using the Morphological Urban Areas (MUA) dataset (Taubenböck et al. 2019), which defines urban areas based on continuous built-up morphology. This ensures consistency across countries with different administrative definitions and aligns with the study's focus on land-cover-driven fragmentation.

All urban areas intersecting mangrove extent were included, resulting in 21 study areas. A uniform 5-km buffer was applied around each Morphological Urban Area to define the analytical frame for urban-associated mangroves. The buffer was selected a priori to capture mangrove patches within and near built-up coastal landscapes using the same rule across all countries. This distance is supported by urban vegetation research that used 5-, 10-, and 15-km buffers around urban cores to examine the effects of urbanization on surrounding vegetation. In that study, vegetation disturbance decreased with distance from the urban core, indicating that the 5-km band represents the nearest peripheral zone of urban influence (Zhang et al. 2020). The distance is also supported by mangrove coastalscape research comparing 500 m, 1 km, and 5 km analytical windows and detecting several mangrove configuration differences at 5 km that were not detected at smaller windows, including differences in cohesion and core-area metrics (Blanco-Libreros and Ramírez-Ruiz 2021). The fixed-distance rule provides a reproducible study frame and avoids inconsistencies arising from differing administrative boundaries across countries. Results are interpreted as conditional on this operational definition of urban-associated mangroves.

Transboundary urban areas represented as a single morphological urban system, such as Singapore–Johor Bahru, were analyzed as joint systems to preserve their spatial continuity and shared coastal

context. All spatial analyses were conducted using the Asia South Albers Equal Area Conic projection to ensure accurate area-based measurements across the study region.

Landscape metrics

Mangrove fragmentation was quantified using four standard patch-level landscape metrics: patch area (PA), perimeter–area ratio (PAR), shape index (SHAPE), and Euclidean nearest-neighbor distance (ENN). These metrics are widely used in landscape ecology and spatial pattern analysis to describe complementary dimensions of patch configuration, including patch size, edge exposure, geometric complexity, and spatial isolation (McGarigal and Marks 1995; Turner and Gardner 2015).

Patch area (PA) measures the total area of each mangrove patch. Perimeter–area ratio (PAR) measures the ratio of edge length to area, with higher values indicating greater edge exposure. Shape index (SHAPE) measures geometric complexity relative to a compact shape, with higher values indicating more irregular boundaries. Euclidean nearest-neighbor distance (ENN) measures the shortest straight-line distance to the nearest-neighboring patch and represents spatial isolation. Metrics were calculated for each patch at each time step (1996, 2007, 2015, and 2020) using ArcGIS Pro.

Potential redundancy among metrics was evaluated using Pearson correlation coefficients for baseline (1996) and endline (2020) conditions. Correlations were weak to moderate in both periods, with no evidence of strong collinearity. This indicates that each metric captures a distinct aspect of patch configuration.

To describe temporal change, two lineage-derived descriptors were also calculated: retention ratio and centroid shift distance. Retention ratio is the proportion of a parent patch's area retained in the corresponding child patch, capturing persistence. The centroid shift distance is the Euclidean distance between the parent and child centroids, capturing spatial displacement. Together, these descriptors quantify how patches change in both structure and position across time. These variables were used as predictors in subsequent analyses.

Before modeling, all variables were expressed as proportional changes between time steps. Predictors

were log- or logit-transformed to stabilize variance, reduce the influence of extreme values, and improve model performance. These transformations ensure comparability across variables and support stable estimation.

Definition of fragmentation typologies

The patch-level mangrove fragmentation typology framework provides a structured representation of how mangrove patches change across time based on recurring spatiotemporal patterns in patch geometry, configuration, area retention, and lineage position (Fig. 3). Four primary fragmentation typologies were defined in this study based on recurring patterns observed in satellite imagery and corresponding changes in landscape metrics: Nibbling, Clearing, Shattering, and Displacing. Nibbling represents gradual edge loss that reduces patch area while largely maintaining patch structure. Clearing refers to large, contiguous removal of mangrove area, often resulting in substantial patch loss. Shattering refers to the subdivision of a patch into multiple smaller fragments, thereby increasing isolation and edge exposure. Displacing captures spatial shifts in patch location, where mangrove area is lost in one location and gained in another, resulting in centroid movement.

In addition to these fragmentation pathways, three outcome classes were included to represent patch states across time steps: Expanding, Stabilizing, and Lost. Expanding refers to increases in patch area. Stabilizing refers to patches with minimal net area change between consecutive time steps. Patches were labeled as Stabilizing when absolute area change was less than 0.1 ha after preprocessing and visual interpretation showed no clear fragmentation, expansion, displacement, or loss. This threshold matches the minimum patch-size threshold and separates low-change patches from patches with measurable spatiotemporal transformation. Lost denotes complete disappearance of a patch between time steps.

These typologies and outcome classes are interpreted as descriptive categories of spatiotemporal change inferred from patch geometry, metric change, and lineage behavior. They do not represent direct causal mechanisms. Together, they allow patch trajectories to be represented as sequences of spatiotemporal transformations across urban areas and over time. These typologies and outcome classes were

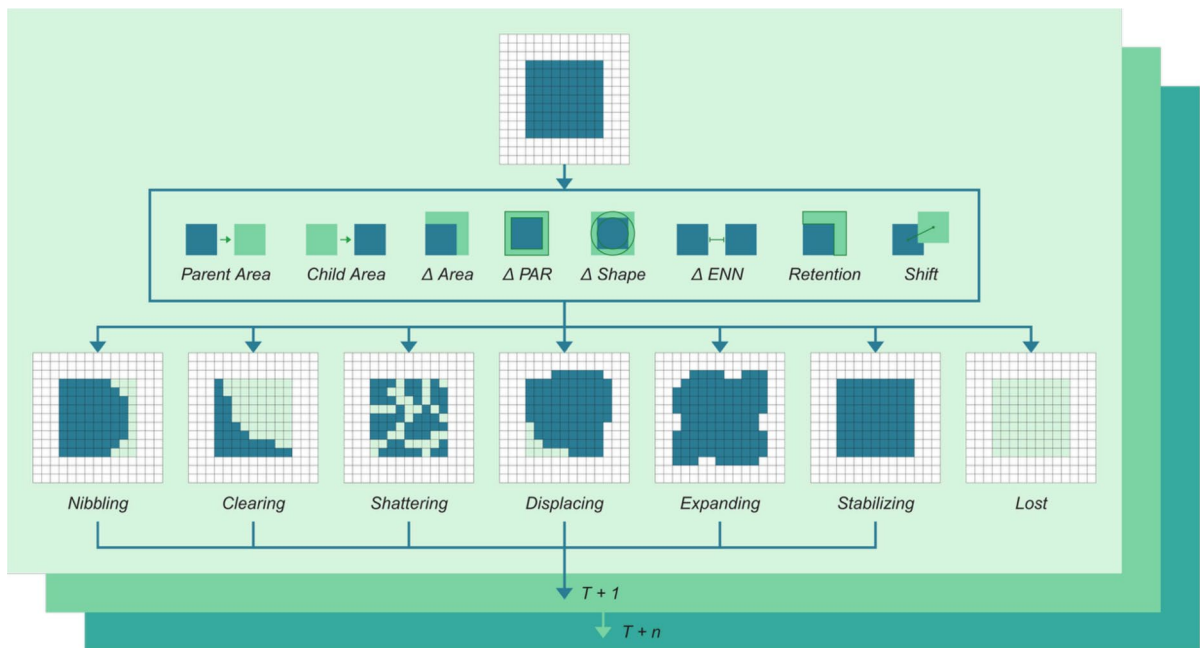


Fig. 3 Patch-level mangrove fragmentation typology framework

developed through iterative visual interpretation of spatial change across multiple urban areas and are not adapted from existing classification frameworks.

Patch lineage reconstruction for trajectory and path-dependence analysis

Fragmentation dynamics were analyzed by reconstructing patch trajectories across consecutive time steps (1996–2007, 2007–2015, and 2015–2020) using a maximum-overlap parent–child tracking framework. Patches were linked across time based on spatial overlap, allowing changes to be traced as continuous sequences.

For each patch at time t (child), spatial overlap was calculated with all patches at time $t-1$ (candidate parents). Each child patch was linked to the parent with the largest overlap, and only this maximum-overlap pair was retained. This rule ensures consistent and reproducible lineage assignment. If overlaps were equal, the larger parent patch was selected.

This framework captures split events in which one parent produces multiple child patches. Merge events were simplified by assigning each child to the dominant parent. Merge events were relatively uncommon. This simplification, therefore, has minimal influence

on the overall trajectory structure while maintaining consistency across time steps. Each linked sequence forms a trajectory describing how a patch changes over time. Patches with no correspondence in the next time step were classified as Lost.

All patches were included in descriptive analyses. However, trajectory and path-dependence analyses were restricted to a closed cohort of patch trajectories with baseline traceability. This cohort includes patches present at the 1996 baseline that could be assigned typology states across the three observation intervals. This design keeps the trajectory length consistent and reduces ambiguity caused by newly appearing patches. Patch loss is represented in this sequence as a baseline patch that no longer has a corresponding patch at a later time step. Results therefore describe baseline-traceable trajectories and do not represent full landscape turnover.

Random forest classification of fragmentation typologies

A Random Forest (RF) model was developed in R to classify fragmentation typologies based on changes in landscape metrics between consecutive time steps.

Each patch was assigned a typology using both its geometric properties and its temporal change.

The model was trained using 1,212 manually classified patches from five urban areas: Cebu–Mactan–Lapu-Lapu (Philippines), Chon Buri (Thailand), Denpasar (Indonesia), Haiphong (Vietnam), and Kuching (Malaysia). These sites were selected to capture a range of urban forms and fragmentation patterns. Typologies were assigned through visual interpretation supported by metric changes.

Eight predictor variables were used, including transformed measures of parent and child area, centroid shift, retention ratio, and changes in PA, PAR, SHAPE, and ENN. These predictors capture both patch structure and change across time.

Although the typologies can be described conceptually using changes in patch geometry and spatial configuration, assigning them consistently across all patch transitions required a multivariate classification approach. No single landscape metric can reliably distinguish among the typologies because similar metric changes can arise from different spatiotemporal transformations. For example, area loss can reflect Nibbling, Clearing, or Shattering depending on whether it occurs through edge retreat, contiguous removal, or subdivision into smaller fragments. Similarly, centroid shift, retention ratio, edge exposure, and isolation may change together in different combinations across urban areas. A rule-based approach would require multiple fixed thresholds and could be sensitive to patch size, mapping uncertainty, and local landscape context. The Random Forest classifier was therefore used to learn recurring combinations of metric change from manually interpreted examples and apply the same classification logic across all study areas. This approach supports reproducible assignment of typologies and enables evaluation of model performance through both internal and external validation.

To reduce spatial and class imbalance, the dataset was split into training (70%) and testing (30%) sets using stratified sampling within each urban area $\times$ typology group. The Random Forest model was trained using 500 trees. This algorithm was selected because it can model non-linear relationships, accommodate correlated predictors, and produce stable classifications across heterogeneous patch transitions.

Fragmentation typologies are interpreted as spatiotemporal process signatures inferred from patch

geometry, metric change, and lineage behavior. They describe observable patterns of change and approximate process-consistent transformations. These classifications provide a consistent basis for comparing fragmentation dynamics across urban areas and should be interpreted as observable signatures, since underlying drivers are not directly observed.

Model performance was evaluated using a confusion matrix, overall accuracy, Cohen's Kappa, confidence intervals, the No Information Rate (NIR), and a binomial test comparing accuracy to the NIR. Model generalizability was assessed using external validation with five additional urban areas. The trained classifier was then applied across all study areas. All models were implemented in R (version 4.5.2) using the 'randomForest' package with default parameters and a consistent interpretation protocol.

Inter-urban trends and typology composition

Inter-urban fragmentation trends were quantified by calculating changes in the mean values of PA, PAR, SHAPE, and ENN between 1996 and 2020. These metrics describe the direction and magnitude of fragmentation across urban areas. To interpret the pathways underlying these trends, fragmentation typologies were summarized using two measures. Proportional composition was calculated as the percentage of patches assigned to each typology. Absolute spatial extent was calculated as the total mangrove area (ha) associated with each typology. Proportional composition describes how fragmentation pathways are distributed, while absolute extent describes how much area they affect. Together, these measures provide a more complete representation of fragmentation dynamics across urban areas.

Quantifying path dependence in fragmentation trajectories

Path dependence in mangrove fragmentation was evaluated using an information-theoretic framework applied to patch-level fragmentation typology trajectories (Nowosad & Stepinski 2019). The analysis tests whether earlier fragmentation states add information about later outcomes beyond the immediately preceding state.

Fragmentation typologies were treated as discrete states observed across three consecutive intervals

(1996–2007, 2007–2015, and 2015–2020). This temporal structure represents the minimum consistent structure required to evaluate multi-step fragmentation trajectories and path dependence at the scale of observation intervals. Each mangrove patch was assigned a three-step trajectory describing its fragmentation history across these intervals. Each trajectory, therefore, represents the sequence of typologies experienced by a patch over time.

Analyses were restricted to a closed cohort of 3,186 mangrove patches that could be assigned complete baseline-traceable trajectories beginning in 1996. Newly appearing patches were excluded because they lack sufficient temporal information to define multi-step trajectories. The cohort therefore follows baseline patches through later states, including cases where a patch disappears during the observation period. This allows patch loss to be analyzed as an outcome of a trajectory rather than as a separate sample-selection rule.

Because observation intervals differ in length, fragmentation typologies represent changes accumulated within each interval. Accordingly, transition probabilities and entropy-based measures describe dependence across these intervals, with outcomes conditioned on states from preceding interval(s).

Path-dependence analyses were conducted on a subset of ten urban areas. These areas were selected based on a minimum threshold of baseline-traceable patch counts required to support stable estimation of multi-step transition probabilities. Urban areas with insufficient numbers of baseline-traceable trajectories do not provide adequate trajectory depth or state-space coverage for reliable estimation of entropy-based dependence measures. This restriction ensures that inference is based on datasets with sufficient statistical support and is not driven by sparse or incomplete trajectory information.

Path dependence was evaluated as a hypothesis test. The null hypothesis (H_0) assumes that the earlier fragmentation state does not improve explanation of the final outcome once the immediately preceding state is known. The alternative hypothesis (H_1) assumes that the earlier fragmentation state adds information about the final outcome.

Two models were estimated. The first-order baseline model assumes that final fragmentation outcomes depend on the immediately preceding typology state. The multi-step history-conditioned model

allows outcomes to depend on both the immediately preceding state and the state from earlier time steps. Operationally, the final interval outcome (2015–2020) was modeled conditional on the preceding typology state (2007–2015) in the first-order baseline model, and conditional on the sequence of preceding typology states (1996–2007 and 2007–2015) in the multi-step history-conditioned model. Model fit was evaluated using log-likelihoods derived from empirical conditional probability distributions. Evidence for path dependence was assessed using likelihood-ratio statistics.

The strength of path dependence was quantified using entropy-based effect sizes. Conditional entropy was calculated for final outcomes given the immediately preceding state and given both prior states. Reductions in conditional entropy (ΔH , in bits) represent how much the earlier state, added to the immediately preceding state, reduces uncertainty about subsequent patch outcomes.

Transition probabilities describe how often specific pathways occur, but they do not quantify how informative prior states are for predicting outcomes. The entropy-based framework addresses this by directly measuring the reduction in uncertainty after adding the earlier fragmentation state. This information-theoretic measure is used here to complement trajectory-based analyses by quantifying the strength of historical constraint, while the primary analysis focuses on identifying recurrent fragmentation pathways and their association with patch loss.

Within this framework, path dependence is defined as a reduction in uncertainty about future outcomes when multi-step prior states are known. Larger reductions in entropy indicate stronger historical constraint on patch outcomes.

Uncertainty estimation and urban-area-level heterogeneity

Because patch-level observations within the same urban area may not be statistically independent—arising from shared environmental conditions, land-use dynamics, and governance processes—uncertainty in path-dependence estimates was quantified using an urban-area-blocked bootstrap. Such dependence is well documented in spatial ecological data, where underlying spatially structured processes can induce correlations among observations and violate the

assumption of independence (Dormann et al. 2007; Zuur et al. 2009). In each bootstrap iteration, entire urban areas were resampled with replacement, and all associated patch trajectories within selected urban areas were retained. This procedure preserves within-area dependence and avoids treating individual patches as independent observations.

Likelihood ratios and entropy reductions were recalculated for each iteration, and percentile-based 95% confidence intervals were derived from the resulting distributions. These intervals quantify the uncertainty associated with path-dependence estimates under spatially structured resampling.

To evaluate sensitivity to differences in patch counts across urban areas and to clarify the unit of inference, path-dependence metrics were summarized using two complementary approaches. First, pooled patch-weighted summaries were calculated, where urban areas contribute in proportion to their number of patch trajectories. Second, urban-area-weighted summaries were calculated, where each urban-area unit contributed equally regardless of patch count. Here, “urban-area-weighted” refers to equal weighting across the study’s urban-area units, including multi-city or transboundary units such as Singapore–Johor Bahru. This comparison checks whether overall results are driven by patch-rich urban areas.

This comparison serves as a robustness check. In spatial ecological analyses, unequal representation across sampling units can bias aggregated results and lead to different interpretations if not explicitly accounted for. Prior studies show that inferred patterns and effect sizes can depend on how data are grouped, aggregated, or scaled, even when underlying processes are similar (Chase and Knight 2013; Murphy 2021; Markham et al. 2023). Evaluating results under alternative aggregation schemes therefore helps ensure that observed patterns are stable and reflect underlying relationships, rather than artifacts of data structure or analytical scale.

Path-dependence metrics—including likelihood ratios, entropy reductions (ΔH), and final-interval probabilities of patch loss—were also calculated separately for each urban area. These urban-area-level estimates characterize variation in historical constraint across different urban contexts. They were interpreted descriptively within the closed cohort of baseline-traceable patch trajectories and were not subjected to formal hypothesis testing.

Path-dependent loss probabilities and recurrent trajectories

Patch-loss probabilities and recurrent trajectories were estimated from sequences of fragmentation-typology transitions along reconstructed patch trajectories. For patches within the closed cohort of baseline-traceable trajectories, prior-history groups were defined using typology states in the 1996–2007 and 2007–2015 intervals. For each prior-history group, the empirical probability of patch loss in the final interval (2015–2020) was calculated.

Full three-step typology trajectories were also identified by grouping patches with the same sequence of typology transitions across all three intervals. Their frequencies were calculated across all baseline-traceable patch trajectories. This full trajectory summary retained loss-ending pathways because patch loss is a key outcome of fragmentation dynamics.

These two summaries serve different purposes. The prior-history loss-probability analysis estimates the observed probability of final-interval loss for each earlier typology history. The full trajectory-frequency analysis identifies dominant pathways, including pathways that end in loss. Both analyses were interpreted descriptively to characterize common fragmentation histories and to highlight pathways associated with higher observed probabilities of patch loss.

Because these trajectories are derived from a closed cohort of baseline patches with traceable typology states, they reflect fragmentation dynamics conditional on baseline inclusion and complete trajectory assignment. They do not capture newly appearing patches or full landscape turnover. Interpretation is therefore limited to observed trajectories within the study and does not assume independence among patches or extend beyond the analyzed contexts.

Robustness and sensitivity analyses

A series of robustness checks was conducted to evaluate the sensitivity of path-dependence results to alternative sample definitions and potential sources of bias.

Analyses were repeated under three conditions. First, patches classified as Lost in earlier intervals were excluded to assess whether terminal Lost states influenced inferred trajectory dependence. Second,

rare fragmentation histories were removed by applying a minimum frequency threshold (≥ 5) to evaluate the influence of infrequent pathways. Third, patches intersecting the analysis boundary were excluded to assess potential edge effects. Each condition isolates a specific source of potential bias, including deterministic terminal states, low-frequency pathways, and boundary-related artifacts.

For each specification, path dependence was re-estimated using the same urban-area-blocked inference framework. This ensures that any differences in results arise from changes in the data and sample definition, while the analytical approach remains consistent.

Results

Trends in mangrove fragmentation (1996–2020)

Across the 21 Southeast Asian urban areas, changes in mean patch area (PA), perimeter–area ratio (PAR), shape index (SHAPE), and Euclidean nearest-neighbor distance (ENN) indicate a consistent shift toward more fragmented mangrove patch configurations

between 1996 and 2020 (Fig. 4). Mean PA declined in 20 of 21 urban areas (median change = -61.68 ha; range: -470.69 to $+4.15$ ha), reflecting widespread reductions in typical patch size. Over the same period, mean PAR increased in 20 of 21 urban areas (median change = $+204.01$; range: -96.94 to $+324.06$), indicating greater edge exposure per unit area. Mean SHAPE also declined in 20 of 21 urban areas (median change = -0.86 ; range: -3.01 to $+0.15$), suggesting that mangrove patches generally became less geometrically complex. Mean ENN increased in 17 of 21 urban areas (median change: $+14.99$; range: -28.14 to $+112.02$), indicating greater spatial isolation in most urban contexts. Taken together, these metrics show that fragmentation intensified across most urban areas through simultaneous changes in patch size, edge exposure, geometry, and spatial separation.

The magnitude of change in fragmentation varied considerably among urban areas. The largest declines in mean PA were observed in Medan (-470.69 ha) and Denpasar (-427.90 ha), followed by Kuching (-273.16 ha), Chon Buri (-221.75 ha), and Sandakan (-201.65 ha). Increases in PAR were strongest in Zamboanga ($+324.06$), Medan ($+314.21$), Bangkok ($+287.16$), Chon Buri ($+262.71$), and



Fig. 4 Trends in mangrove fragmentation in Southeast Asian urban areas. **a** Change in mean patch area (ha); **b** Change in mean perimeter–area ratio (m/ha); **c** Change in mean shape index ($m/\sqrt{m^2}$); **d** Change in mean Euclidean nearest-neighbor distance (m)

Denpasar (+259.86). Declines in SHAPE were concentrated in Medan (−3.01), Denpasar (−1.95), Surabaya (−1.94), Chon Buri (−1.87), and Zamboanga (−1.85), indicating that increases in edge exposure were often accompanied by simplification of patch geometry. In contrast, isolation-driven fragmentation was most pronounced in Manado (+112.02 ENN), followed by Zamboanga (+50.51) and Makassar (+49.37). These patterns show that fragmentation varies across urban areas in both magnitude and dominant form, reflecting different combinations of spatiotemporal change pathways.

Manado represented a notable exception to regional trends for several metrics. It showed a small increase in mean PA (+4.15 ha), a decrease in PAR (−96.94), and a slight increase in SHAPE (+0.15), while still exhibiting a substantial increase in ENN (+112.02). This pattern indicates that fragmentation in some urban areas can be driven primarily by increasing isolation, even when patch size and shape remain relatively stable.

Decreases in ENN were otherwise limited to Basilan (−19.27), Cebu–Mactan–Lapu–Lapu (−11.33), Cotabato (−28.14), and Haiphong (−3.37), despite these urban areas generally following broader patterns of declining PA and SHAPE and increasing PAR. This shows that fragmentation does not always involve increasing isolation and may reflect local processes that maintain or reduce distances between patches.

Overall, mangrove fragmentation intensified over the study period (1996–2020) across most Southeast Asian urban areas, with substantial variation in the pathways shaping patch structure and spatial configuration.

Random forest model performance and validation

The Random Forest model achieved an overall accuracy of 0.82 and a Cohen's kappa of 0.75 (Table 1), indicating strong classification performance. The 95% confidence interval for accuracy ranged from 0.78 to 0.86. Model accuracy was significantly higher than the No Information Rate (0.45; $p < 0.001$), showing that the model performs well above random or majority-class classification.

Model generalizability was further supported by independent external validation (Table 2) using manually classified patch samples from five urban areas

Table 1 Performance of the random forest model

Metric	Value
Accuracy (Acc)	0.8242
Kappa	0.7537
95% Confidence Interval (CI)	[0.7811, 0.8619]
No Information Rate (NIR)	0.4533
P-value [Acc > NIR]	2.9038e-48

not included in model training: Cotabato, Sandakan, Singapore–Johor Bahru, Surabaya, and Vung Tau. Across these sites, classification accuracy ranged from 0.84 to 0.92, with kappa values between 0.78 and 0.88. In all cases, accuracy significantly exceeded the corresponding No Information Rates ($p < 0.001$), and confidence intervals indicated consistently high reliability. These outcomes show that the model performs consistently across urban areas with different fragmentation patterns and maintains strong predictive accuracy outside the training dataset.

Together, these findings demonstrate that the classification framework is both accurate and transferable across diverse urban contexts. However, classification errors remain present, and typology assignments should be interpreted as probabilistic representations of spatiotemporal change signatures. Individual patch labels therefore contain some degree of uncertainty, even when overall model performance is high.

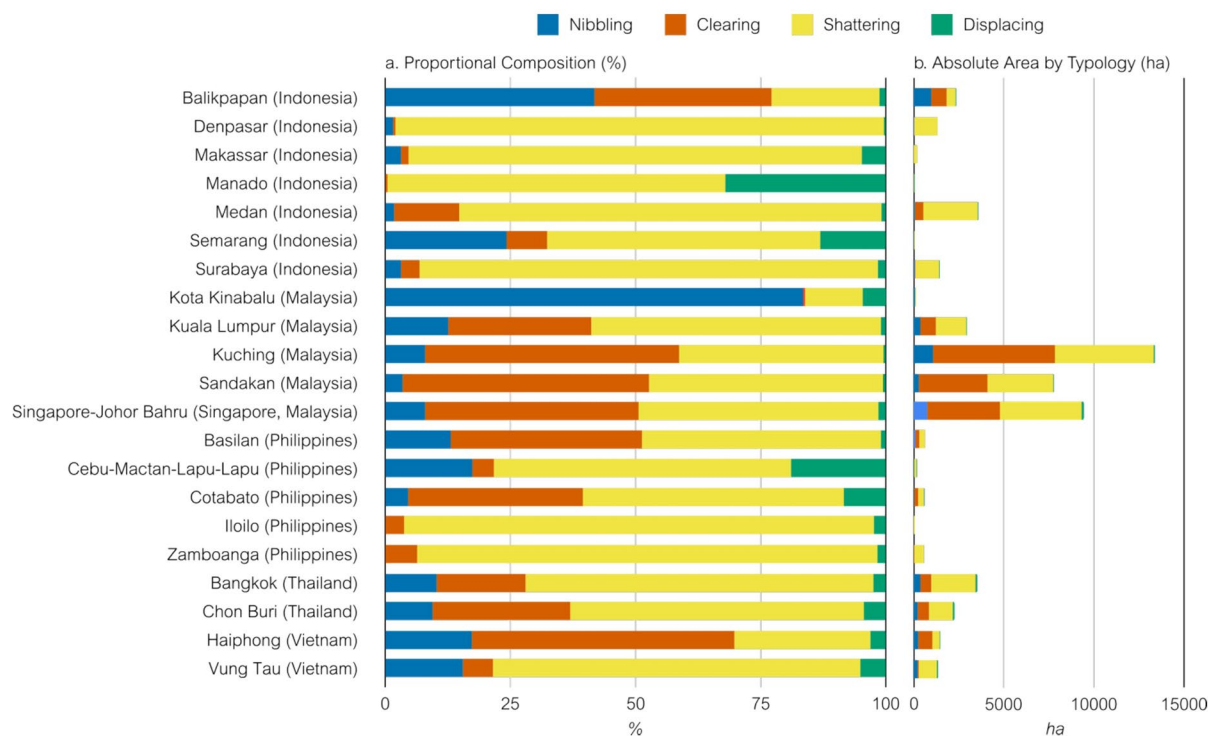
Inter-urban composition of mangrove fragmentation typologies

Fragmentation typology composition in 2020 varied widely across urban areas, showing strong differences in both the relative dominance and spatial extent of fragmentation pathways (Fig. 5). When expressed as proportions, Shattering was the dominant typology in most urban areas, accounting for more than 80% of fragmented patches in Denpasar, Makassar, Surabaya, Iloilo, and Zamboanga. In contrast, some urban areas showed distinct profiles. For example, in Kota Kinabalu, Nibbling accounted for 83.4% of fragmented patches. These differences show that fragmentation pathways vary across urban areas and reflect diverse spatiotemporal change patterns and local landscape conditions.

Several urban areas showed more mixed typology compositions, with substantial contributions from

Table 2 External independent validation

Urban area	Accuracy (Acc)	Kappa	95% CI	NIR	P-value [Acc > NIR]
Cotabato (PH)	0.85	0.7793	[0.7647, 0.9135]	0.37	<2.2e-16
Sandakan (MY)	0.85	0.8039	[0.7647, 0.9135]	0.28	<2.2e-16
Singapore–Johor Bahru (SG, MY)	0.86	0.7829	[0.7763, 0.9213]	0.54	9.974e-12
Surabaya (ID)	0.92	0.8783	[0.8484, 0.9648]	0.47	<2.2e-16
Vung Tau (VN)	0.84	0.7781	[0.7746, 0.8897]	0.40	<2.2e-16

**Fig. 5** Composition of mangrove fragmentation typologies in Southeast Asian urban areas (1996–2020). **a** Proportional composition (%); **b** Absolute area by typology (ha)

multiple pathways. Kuching, Sandakan, Haiphong, and Singapore–Johor Bahru exhibited relatively balanced proportions of Clearing and Shattering. Balikpapan and Cotabato showed notable contributions from Nibbling alongside Clearing. Across all urban areas, Displacing consistently represented a small share of fragmentation, rarely exceeding 5–10%. These patterns indicate that fragmentation often involves multiple transformation pathways occurring within the same urban landscape.

Patterns based on absolute fragmented area revealed additional contrasts that were not visible

from proportional composition alone. Large metropolitan areas—such as Kuching, Singapore–Johor Bahru, Sandakan, and Medan—contained the greatest total area of fragmented mangroves, with extensive areas classified as Shattering and Clearing. In contrast, smaller urban areas, including Iloilo, Kota Kinabalu, and Manado, had limited total fragmented area despite a strong proportional dominance of specific typologies.

Representative patch-level maps further illustrate how these typologies manifest in local spatial contexts (Fig. 6). Nibbling is characterized by edge retreat

along an existing mangrove patch. Clearing appears as contiguous removal of mangrove area. Shattering appears as subdivision of a larger patch into smaller fragments. Displacing appears as spatial relocation of mangrove area across consecutive time steps. These examples complement the aggregate typology results by showing the mapped forms behind the classified patch transitions.

Proportional composition describes how fragmentation pathways are distributed within an urban area, while absolute area describes how much space they occupy. Considering both measures together provides a more complete understanding of fragmentation dynamics across urban areas.

In several urban areas, including Surabaya, Cebu–Mactan–Lapu-Lapu, and Zamboanga, patches classified as Expanding accounted for a measurable share of the total mangrove area. This shows that fragmentation and expansion can occur simultaneously within a single urban system. This coexistence reflects ongoing spatial reconfiguration, where loss, subdivision, and growth processes operate concurrently across the landscape.

Inter-urban variation in path dependence of mangrove fragmentation

Inter-urban variation in path dependence across baseline-traceable mangrove patch trajectories is illustrated using dot–range plots ordered by decreasing memory strength (ΔH ; Fig. 7). To complement the trajectory-based analysis of fragmentation pathways, we quantify the strength of historical constraint on patch outcomes using entropy-based measures. Across the ten urban areas analyzed, entropy-based effect sizes ranged from 0.148 to 0.438 bits, indicating variation in the extent to which earlier fragmentation states reduce uncertainty about later patch outcomes.

Urban areas such as Cotabato, Basilan, and Balikpapan exhibited the strongest historical constraint ($\Delta H > 0.40$ bits), whereas Bangkok showed consistently lower values ($\Delta H \approx 0.15$ bits). These differences indicate that the influence of fragmentation history on patch outcomes varies across urban contexts and is associated with variation in fragmentation pathways. These results quantify how prior states constrain patch outcomes within a closed cohort of

baseline-traceable patch trajectories across different urban systems.

Likelihood ratio statistics generally followed the same ranking as ΔH , providing consistent support for history-conditioned models across urban areas. However, some urban areas with large numbers of baseline-traceable trajectories—such as Bangkok and Kuching—showed high likelihood ratio values despite relatively small entropy reductions. This pattern reflects the influence of sample size on likelihood-based statistics, while entropy reduction reflects the magnitude of historical constraint on patch outcomes. Considering both metrics together helps distinguish statistical support for path dependence from the strength of its effect.

Final-interval patch-loss probabilities varied independently of entropy-based effect sizes across urban areas. Urban areas with similar ΔH values often exhibited different loss probabilities. This indicates that the strength of historical constraint is not directly linked to overall patch loss rates.

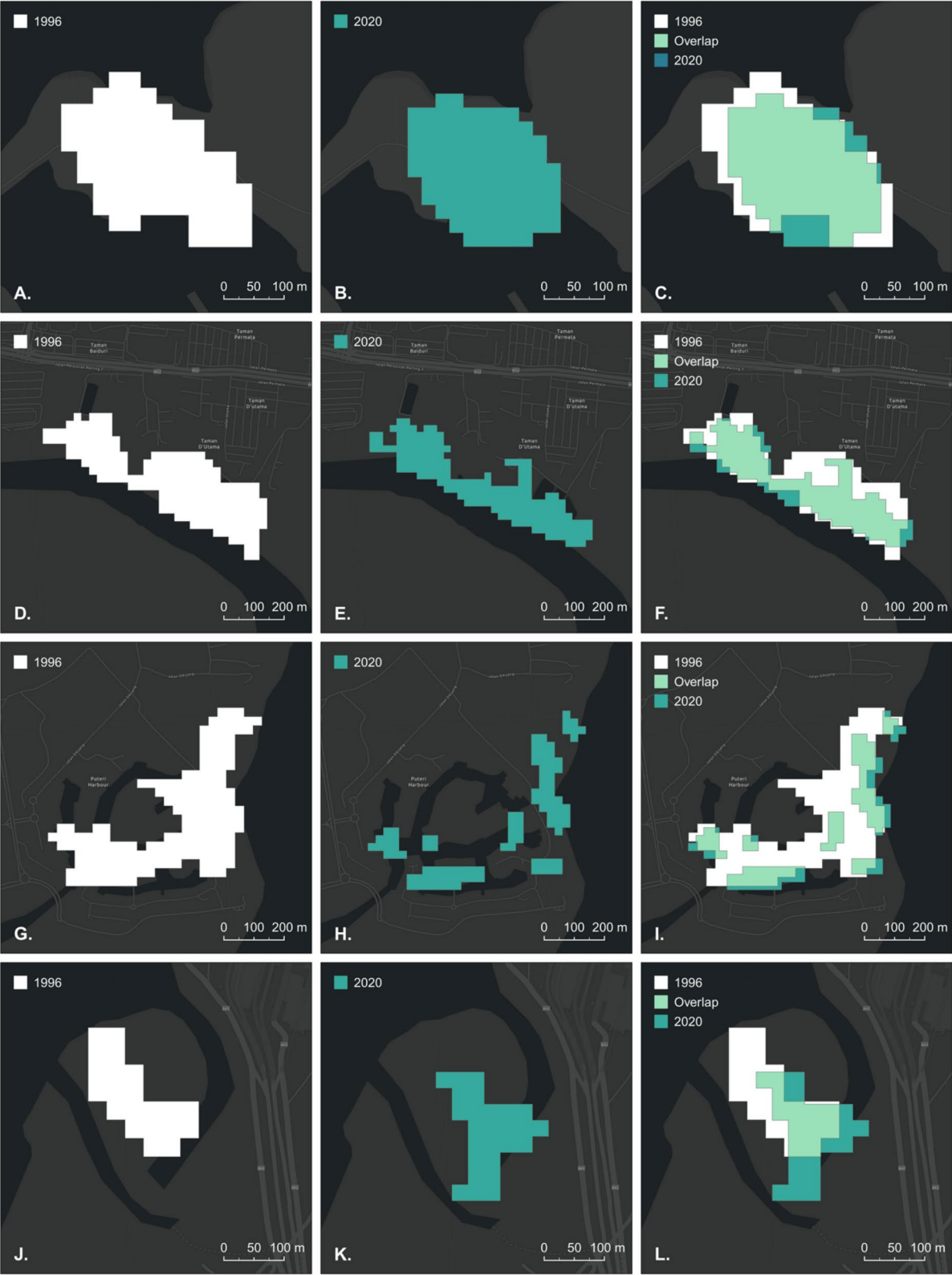
These results indicate that path dependence describes how fragmentation pathways are structured through time, shaping the sequence and likelihood of transitions among typologies. At the same time, aggregate loss rates represent a separate dimension of landscape change that is not fully explained by trajectory structure alone.

Overall, fragmentation history contributes to the organization of patch outcomes across urban areas, but the strength of this effect and its relationship to patch loss vary across contexts.

Path-dependent probability of mangrove patch loss

Conditional probabilities of mangrove patch loss varied across prior fragmentation histories (Fig. 8). Here, conditional probability refers to the observed probability that a patch is classified as Lost in the final interval given its preceding sequence of fragmentation typologies. This analysis links the history of fragmentation to variation in patch loss outcomes.

Loss probabilities were higher for several histories involving Shattering and Clearing, indicating that subdivision and large-area removal were often associated with later patch loss. However, loss probabilities still varied among histories that included these typologies. For example, histories involving Shattering could lead to different outcomes depending on



◀Fig. 6 Representative patch-level examples of mangrove fragmentation typologies. Panels show the 1996 extent, the 2020 extent, and the overlapping area for four classified patch transitions: A–C Nibbling, D–F Clearing, G–I Shattering, and J–L Displacing

the preceding typology sequence. This shows that the most recent state alone does not fully explain patch fate.

Some prior histories include states already classified as Lost before the final interval. These terminal histories were retained in the full trajectory summary because they describe common loss-ending pathways in the baseline-traceable cohort. Their influence was evaluated through the robustness analysis that excluded early Lost trajectories.

These pathway-specific probabilities are descriptive and are not interpreted as formal statistical differences among trajectories. The results show associations between fragmentation histories and loss likelihood across baseline-traceable patch trajectories, with variation across urban areas and among similar sequences. Overall, fragmentation history shapes the distribution of loss probabilities across trajectories, and patch fate remains context-dependent.

Recurrent fragmentation trajectories and loss-terminating sequences

Across all urban areas, a small number of fragmentation trajectories accounted for a large share of observed patch histories among baseline-traceable patches (Fig. 9; Table 3). Figure 9 provides a visual summary of class transitions across the three observation intervals, and Table 3 reports the exact counts for the most prevalent full trajectories. Table 3 summarizes the ten most prevalent fragmentation trajectories across the ten urban areas included in the analysis, accounting for 1,209 of the 3,186 closed-cohort patches analyzed.

The most common trajectory, Shattering→Lost→Lost, was observed for 632 patches, far exceeding all other sequences. Other common trajectories included Clearing→Lost→Lost and Shattering→Shattering→Lost. These same dominant sequences are visible in Fig. 9 as large flows involving Shattering, Clearing, and Lost classes. These patterns indicate that internal subdivision and large-area removal often precede terminal patch loss.

The concentration of patch histories within a small set of trajectories indicates that fragmentation dynamics are organized around recurring pathways. These pathways provide a basis for interpreting fragmentation dynamics and associated loss risk across urban areas and are primarily characterized by repeated subdivision and large-area removal processes.

In addition to these loss-associated trajectories, a smaller group of pathways reflects persistence or partial recovery following fragmentation. Examples include Shattering→Expanding→Expanding and Expanding→Stabilizing→Displacing. These pathways appear as smaller flows in Fig. 9, indicating that non-loss trajectories occurred alongside the more common loss-associated sequences. These trajectories show that fragmentation does not always lead to patch loss and that outcomes can diverge after initial disturbance, indicating that multiple pathways remain possible following fragmentation events.

Across all urban areas, a small number of loss-terminating trajectories accounted for a large share of observed patch losses among baseline-traceable patches (Table 4). Table 4 summarizes the ten most prevalent fragmentation trajectories culminating in patch loss across the ten urban areas included in the analysis, accounting for 1,107 of the 3,186 closed-cohort patches analyzed. Pathways involving Shattering and Clearing are the most common among patches that ultimately disappear, indicating that a subset of fragmentation trajectories is disproportionately associated with loss outcomes.

At the same time, the presence of alternative trajectories indicates that similar initial fragmentation states do not always produce the same outcomes, highlighting variability in how fragmentation pathways unfold across patches and contexts.

Because these trajectories are derived from the closed cohort of baseline patches with traceable trajectories, the reported frequencies emphasize pathways that remain observable across multiple time steps. The cohort includes lineages that later transition to Lost. It should therefore be interpreted as baseline-traceable, rather than persistence-only.

Taken together, the trajectory counts and visualized flows indicate that fragmentation dynamics are structured around recurring sequences. The sequence of spatiotemporal transformations influences how patch outcomes evolve over time.

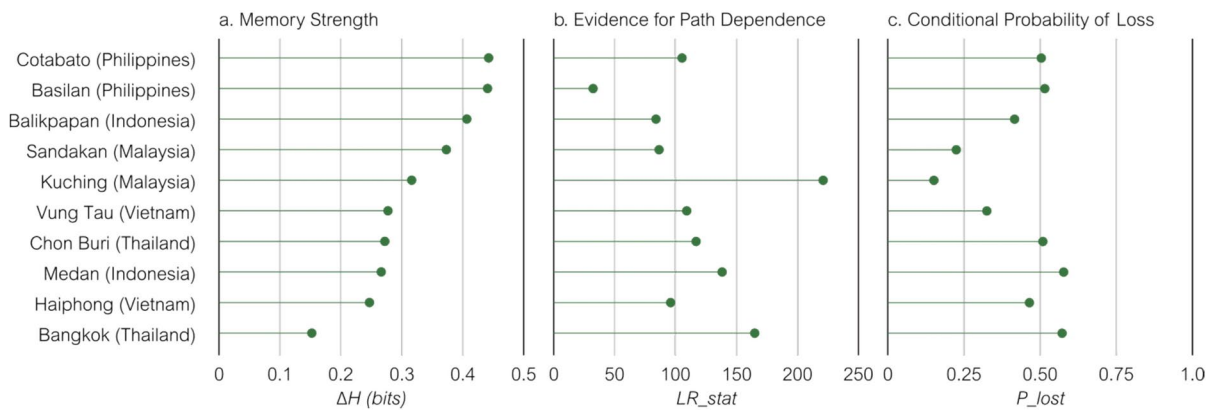


Fig. 7 Path dependence in mangrove fragmentation. **a** Memory strength (ΔH , bits); **b** Likelihood ratio statistic (LR_{stat}); **c** Conditional probability of loss (P_{Lost})

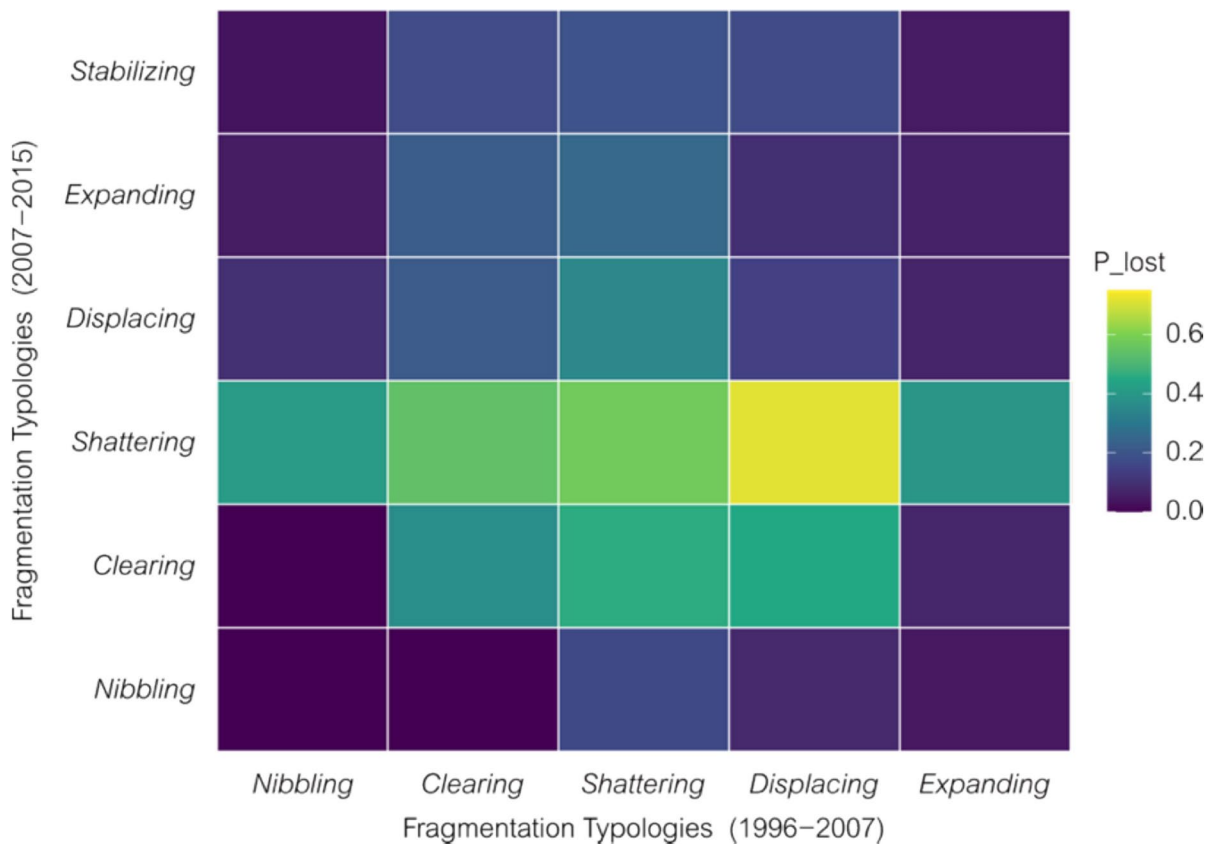


Fig. 8 Conditional probabilities of mangrove patch loss across fragmentation trajectories. Transitions originating from Lost states in 2007–2015 are excluded

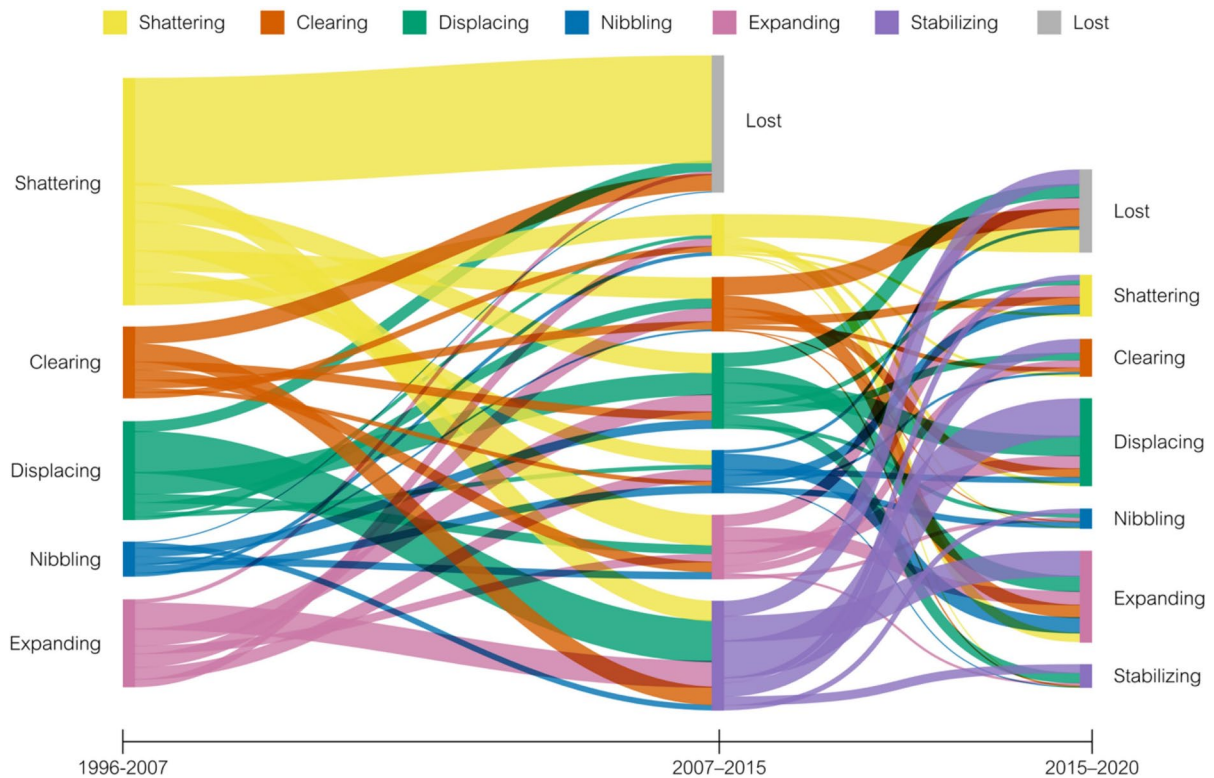


Fig. 9 Sankey diagram of patch-lineage transitions among mangrove fragmentation typologies (1996–2007, 2007–2015, and 2015–2020)

Table 3 Most prevalent mangrove fragmentation trajectories across urban areas (top 10)

Path	Patches (n)
Shattering→Lost→Lost	632
Clearing→Lost→Lost	101
Displacing→Stabilizing→Displacing	96
Shattering→Shattering→Lost	72
Displacing→Lost→Lost	65
Shattering→Clearing→Lost	57
Expanding→Stabilizing→Displacing	52
Shattering→Expanding→Lost	46
Shattering→Expanding→Expanding	44
Shattering→Stabilizing→Expanding	44

Table 4 Most prevalent fragmentation trajectories culminating in patch loss (top 10)

Path	Patches (n)
Shattering→Lost→Lost	632
Clearing→Lost→Lost	101
Shattering→Shattering→Lost	72
Displacing→Lost→Lost	65
Shattering→Clearing→Lost	57
Shattering→Expanding→Lost	46
Displacing→Stabilizing→Lost	43
Shattering→Displacing→Lost	41
Displacing→Clearing→Lost	27
Expanding→Lost→Lost	23

Robustness and stability of path dependence estimates

This section evaluates whether evidence of path dependence is consistent across different aggregation

and sampling approaches. These analyses provide a complementary assessment of the stability of historical constraint within the broader trajectory-based framework. Across all urban areas, pooled analyses of baseline-traceable mangrove patch trajectories were

Table 5 Summary of pooled and urban-area-weighted path dependence results

Urban-area-weighted estimates give equal weight to each urban-area unit, regardless of patch count. This weighting reflects the study's analytical units, which include multi-city and transboundary urban areas

Metric	Pooled (Patch-Weighted)	Urban-Area-Weighted
Number of units (n)	3186 patches	10 urban areas
Log-likelihood (baseline, no history)	-4010.9	–
Log-likelihood (history-conditioned)	-3760.9	–
Likelihood ratio statistic	500.08	–
Conditional entropy, $H(Y \text{prev})$ (bits)	1.816	–
Conditional entropy, $H(Y \text{prev}, \text{prev2})$ (bits)	1.703	–
ΔH (entropy reduction, bits)	0.113	0.130
Bootstrap mean ΔH	0.164	0.192
95% urban-area-blocked bootstrap CI	[0.120, 0.231]	[0.142, 0.259]

used to evaluate path dependence (Table 5). These analyses provide consistent evidence that fragmentation trajectories are influenced by prior states.

Models that conditioned patch outcomes on multi-step prior fragmentation states showed improved fit compared to first-order baseline models, yielding a likelihood ratio statistic of 500.08. The corresponding entropy reduction ($\Delta H = 0.113$ bits) indicates that information about the earlier fragmentation state, beyond the immediately preceding state, reduces uncertainty in subsequent patch outcomes within the closed cohort of patches traceable from 1996. This result shows that fragmentation history provides measurable predictive information beyond the current state alone.

Entropy reductions were consistently positive across all analyses, indicating that the contribution of fragmentation history to uncertainty reduction is stable. Urban-area-blocked bootstrap resampling produced a higher mean ΔH (0.164 bits; 95% CI: 0.120–0.231 bits), indicating that the observed reduction in uncertainty is not driven by any single urban area.

Conditional entropy declined from 1.816 bits when conditioning on one prior state to 1.703 bits when conditioning on two prior states. This shows

that incorporating additional historical information further reduces uncertainty in patch outcomes. Together, these findings indicate that fragmentation dynamics reflect multi-step trajectories, with earlier states continuing to influence later outcomes.

To assess sensitivity to differences in patch counts across urban areas, path dependence was also summarized using an urban-area-weighted approach, in which each urban-area unit contributed equally regardless of patch count. Under this specification, entropy reduction remained positive and slightly larger ($\Delta H = 0.130$ bits; bootstrap mean = 0.192 bits; 95% CI: 0.142–0.259 bits). This indicates that the observed historical dependence is not driven by a small number of patch-rich urban areas.

The consistency between pooled and urban-area-weighted estimates shows that path dependence is detectable across urban contexts and is not an artifact of uneven sampling.

Additional robustness checks—including the exclusion of early-loss patches, the trimming of rare trajectories, and the removal of boundary-intersecting patches—produced comparable entropy reductions across alternative specifications (Table 6). This consistency shows that the detection of path dependence

Table 6 Robustness checks for path dependence in mangrove fragmentation dynamics

Robustness check	Urban areas (n)	Patches (n)	Likelihood ratio statistic	ΔH (entropy reduction, bits)
Primary (full typologies; urban area-blocked inference)	10	3186	500.08	0.113
Exclude early Lost	10	2359	500.08	0.153
Trim rare histories within urban areas	10	2882	505.09	0.126
Exclude boundary-intersecting patches	10	2965	488.48	0.119

is stable across different sample definitions and modeling assumptions.

Although the magnitude of ΔH is modest, its consistent detection across pooled, weighted, and sensitivity analyses indicates a stable and non-random historical signal. Taken together, these findings provide consistent evidence that fragmentation trajectories exhibit measurable, though modest, historical dependence that persists across analytical specifications.

Discussion

Fragmentation as a historically structured process in urban mangrove landscapes

This study shows that mangrove fragmentation in urban landscapes in Southeast Asia unfolds as sequences of spatiotemporal transformations along baseline-traceable patch trajectories at the temporal resolution of the available observations. Across the study urban areas, mangrove patches became smaller, simpler, and more isolated between 1996 and 2020, reflecting broader regional trends of increasing fragmentation under cumulative human pressure (Bryan-Brown et al. 2020). Within this context, the analysis identifies recurrent fragmentation pathways and their association with patch outcomes, while complementary analyses quantify how prior states constrain these pathways.

By integrating patch-level typology classification with trajectory reconstruction, this study shows that similar patterns in commonly used landscape metrics can arise from different spatiotemporal change pathways. This means that comparable metric trends do not necessarily reflect the same underlying fragmentation dynamics. As a result, interpretations based only on static metrics remain limited because they do not capture how spatial change unfolds across time steps. These observations support long-standing critiques that snapshot-based assessments can obscure the pathways through which fragmentation develops (McGarigal & Cushman 2002; Cushman et al. 2008; Fahrig 2013).

Urban mangrove systems provide a particularly strong context for examining historically structured fragmentation. Permanent infrastructure, coastal engineering, and long-term land-use change constrain how patches reorganize, making past spatial

configurations more influential in shaping future patterns. In this setting, fragmentation pathways dominated by Shattering and Clearing are consistently associated with a higher probability of patch loss among baseline-traceable patch trajectories. These pathway-level patterns form the primary basis for interpreting fragmentation dynamics across urban mangrove systems. These pathways reflect internal subdivision and large-area removal, which reduce structural cohesion and increase exposure to further disturbance (Haddad et al. 2015).

Because trajectories were reconstructed from a closed cohort of baseline-traceable patches, the observed patterns reflect dynamics along baseline-origin histories. This allows earlier spatial transformations to be directly linked to later outcomes, including patch loss. It provides a clearer view of how fragmentation accumulates through time.

Together, these findings show that fragmentation is shaped by both spatial configuration and the sequence of transformations that produce it. Viewing fragmentation as a temporally structured process provides a more complete basis for interpreting landscape change in urban mangrove systems.

Inter-urban heterogeneity and the limits of aggregate metrics

Mangrove fragmentation increased across nearly all urban areas, but its magnitude and spatial form varied widely. In some urban areas, fragmentation was primarily associated with increasing patch isolation. In others, it reflected geometric simplification, internal subdivision, or abrupt clearing. These differences show that similar overall trends in fragmentation metrics can emerge from distinct combinations of spatiotemporal change pathways.

Fragmentation typologies also reveal that pathways that are most common in relative terms are not always those affecting the largest total area. Smaller urban areas may show strong proportional dominance of a single typology while affecting limited total area. Larger urban areas may contain extensive fragmented area across multiple typologies even when proportions appear more balanced.

Examining both proportional composition and absolute fragmented area provides complementary insight into how fragmentation is expressed across landscapes. Proportional measures describe how

pathways are distributed within an urban area, while absolute measures reflect their overall spatial extent. Considering both perspectives together helps reveal differences among urban areas that remain hidden when using a single summary measure.

These findings reinforce concerns that commonly used landscape metrics are often closely linked to total habitat area and may obscure underlying fragmentation mechanisms when spatial configuration is not explicitly considered (Rutledge 2003; Wang et al. 2014; Adler and Jedicke 2022). In this study, similar changes in patch-level metrics were associated with different typological compositions, indicating that aggregate metrics alone cannot distinguish among underlying pathways.

By combining typology-based classification with spatial extent, this study provides a more detailed and process-oriented framework for comparing fragmentation dynamics across heterogeneous urban systems. This approach reveals contrasts that are not visible in urban-area-level summaries and supports more accurate interpretation of fragmentation patterns across urban areas (Hadley and Betts 2016; Fardila et al. 2017).

Quantifying path dependence in mangrove fragmentation dynamics

A central contribution of this study is the explicit quantification of path dependence in urban mangrove fragmentation along baseline-traceable patch trajectories. This analysis complements the trajectory-based identification of fragmentation pathways by quantifying the strength of historical constraint on patch outcomes. Using a parent–child maximum-overlap tracking framework, patch trajectories were reconstructed across time steps, allowing fragmentation to be analyzed as sequences of spatiotemporal transformations.

Across major urban mangrove systems, multi-step history-conditioned models consistently outperformed first-order baseline models. This shows that fragmentation dynamics along baseline-traceable trajectories are influenced by earlier transformations. In other words, historical states provide measurable information about future patch outcomes beyond the immediately preceding state. This pattern is consistent with landscape ecological theory on disturbance legacies and ecological memory (Turner 2010; Peters

et al. 2011; Turner and Gardner 2015; Johnstone et al. 2016).

Entropy reductions (ΔH) were modest but consistently positive and robust across sensitivity analyses. ΔH represents the reduction in uncertainty about patch outcomes when the earlier fragmentation state is added to the immediately preceding state. These effect sizes indicate a detectable but limited influence of multi-step prior states on subsequent outcomes. This provides a quantitative complement to trajectory-based findings by indicating how much prior states constrain the pathways through which fragmentation unfolds. Likelihood-ratio statistics capture the cumulative detectability of this dependence across many trajectories, while ΔH provides a scale-independent measure of its strength.

Path dependence varies across urban contexts. In systems with extensive baseline-traceable patch networks, repeated transformations accumulate along trajectories, allowing historical constraint to emerge more clearly. In systems characterized by rapid turnover, fragmentation outcomes are more strongly influenced by short-term disturbances, resulting in weaker detectable dependence.

The additional reduction in conditional entropy when multi-step histories are included shows that patch outcomes reflect cumulative trajectories across time, not only the immediately preceding state. Interpretation is intentionally restricted to a closed cohort of patches traceable from the baseline year. This isolates lineage-conditioned dynamics and ensures that observed path dependence reflects baseline-traceable histories. As a result, these observations describe conditional dynamics among baseline-traceable patches and do not represent system-wide landscape behavior.

Overall, these results provide consistent evidence that fragmentation dynamics are shaped by historical sequences of spatiotemporal change, with prior states contributing measurable, though non-deterministic, influence on future outcomes.

Recurrent trajectories and fragmentation pathways to loss

Mangrove patch loss among baseline-traceable patches was concentrated within a small number of recurring fragmentation trajectories, especially those dominated by Shattering and Clearing. This pattern shows that specific sequences of spatiotemporal

transformations are more strongly associated with degradation outcomes.

These pathways reflect progressive structural changes that reduce patch cohesion and increase exposure to disturbance, consistent with landscape ecological theory on disturbance legacies and structural change (Turner 2010; Johnstone et al. 2016). Over time, repeated subdivision and removal can accumulate along trajectories, increasing vulnerability to eventual loss.

At the same time, some trajectories show stabilization or expansion following earlier fragmentation. These patterns demonstrate that fragmentation does not lead to a single outcome and that patch trajectories can diverge even under similar initial conditions. Persistence and recovery depend on how fragmentation pathways unfold across time, including the degree of subdivision, edge exposure, and spatial arrangement of remaining patch components.

These findings highlight that fragmentation pathways differ in their association with vulnerability and persistence. Understanding these differences requires examining how spatial transformations accumulate across time steps. This trajectory-based perspective provides a clearer basis for identifying high-risk pathways and interpreting fragmentation dynamics beyond single-time spatial summaries. Accordingly, pathway identification serves as the primary analytical lens, while path dependence quantifies the degree to which these pathways are historically constrained.

Methodological contributions to landscape ecology

This study demonstrates how combining machine-learning-based typology classification with information-theoretic analysis can support process-oriented approaches to fragmentation. By reconstructing patch-level trajectories along consistent lineages, the framework captures how spatial transformations unfold and accumulate across time.

Machine learning supports this approach by enabling consistent classification of fragmentation typologies across large and complex datasets. The strong performance and generalizability of the Random Forest classifier show that typology-based analyses can be applied reliably across diverse urban contexts. These typologies approximate spatiotemporal process signatures and provide a structured basis for comparing fragmentation dynamics across landscapes.

Entropy reduction provides a clear and interpretable way to quantify historical constraint. In this framework, entropy-based measures are used as a complementary tool to quantify information gain, alongside trajectory-based analyses that identify dominant fragmentation pathways. It measures how information about multi-step prior fragmentation states reduces uncertainty in future outcomes. This approach links information-theoretic metrics with ecological processes while avoiding assumptions of deterministic behavior.

The use of pooled and urban-area-weighted summaries, along with urban-area-blocked bootstrap resampling, ensures that results are not driven by uneven patch counts or spatial clustering. The urban-area-weighted summary reflects the study's urban-area units and gives equal influence to each unit, including transboundary or multi-city study areas. These design choices strengthen the robustness of the findings and clarify the relationship between analytical units and inference.

Together, these methodological components form a transparent and transferable framework for analyzing fragmentation trajectories and historical contingency in complex, human-modified landscapes.

Limitations

Several limitations should be considered when interpreting these results. First, fragmentation dynamics were evaluated over unequal observation intervals (1996–2007, 2007–2015, and 2015–2020), reflecting data availability and the four analytical time points selected for trajectory reconstruction. As a result, fragmentation typologies and transition probabilities represent changes aggregated within each interval. These findings therefore describe fragmentation across discrete time windows and do not represent time-standardized or annualized rates of change.

Second, trajectory reconstruction and path-dependence analyses were restricted to a closed cohort of mangrove patches that could be assigned baseline-traceable trajectories beginning in 1996. This design isolates historical dynamics along consistent patch histories and avoids confounding path dependence with newly appearing patches or ambiguous lineage assignments. However, it focuses on baseline-traceable patches and does not capture newly emerging patches. As a result, trajectories associated

with newly formed mangrove patches may be under-represented, and path-dependence estimates reflect lineage-conditioned dynamics rather than the full landscape system. These constraints reflect the temporal resolution of the available data and do not affect the identification of broad fragmentation pathways across time.

Third, patch lineage reconstruction used a maximum-overlap parent–child tracking framework that preserves split events but simplifies merge events by assigning child patches to a dominant parent. Although merge events were relatively uncommon, this simplification may underrepresent more complex patch fusion processes and alternative lineage pathways. Consequently, reconstructed trajectories emphasize dominant lineage patterns and may not capture all possible patch genealogies.

Fourth, spatial autocorrelation is an inherent feature of landscape structure, especially in urban environments where infrastructure, land-use patterns, and environmental gradients create spatial clustering. This study does not explicitly model spatial dependence among neighboring patches. Instead, spatial structure is partially addressed through urban-area-blocked bootstrap resampling, which preserves within-area clustering during inference. As a result, the analysis focuses on temporal dependence along patch trajectories, without explicitly capturing spatial spillover effects.

Fifth, raster preprocessing choices and minimum patch-size thresholds may influence observed fragmentation patterns. Majority filtering, raster-to-polygon conversion, and exclusion of patches smaller than 0.1 ha may smooth fine-scale boundaries and reduce representation of small or narrow patches. Although these steps were applied consistently across all time periods and urban areas, they may still influence estimates of patch size, edge complexity, and isolation.

Sixth, urban-associated mangroves were defined using a uniform 5-km buffer around morphologically delineated urban areas. This rule provides a reproducible comparative frame across countries. Results may differ under alternative distance thresholds, functional definitions of urban influence, or site-specific coastal planning boundaries.

Seventh, analyses were restricted to urban areas with sufficient mangrove patches to support trajectory modeling. This limits generalization to larger or more patch-rich urban systems and may not fully represent

fragmentation dynamics in smaller or highly fragmented landscapes. Variation in the strength of path dependence across urban areas further indicates that historical constraint depends on landscape structure and disturbance regimes.

Finally, this study relies on the Global Mangrove Watch dataset, which is provided as raster data with 25-m pixel spacing. Small, narrow, or highly fragmented mangrove patches may be underrepresented, especially those embedded within dense urban infrastructure. This limitation may lead to underestimation of the intensity of fine-scale fragmentation and overestimation of the mean patch size in highly urbanized settings. The use of interpretable machine-learning models improves transparency and reproducibility. Classification outcomes remain sensitive to training data quality and typology definitions.

Together, these limitations indicate that the results should be interpreted as conditional and scale-dependent representations of fragmentation dynamics within the observed data and analytical framework.

Future directions

Future research could extend this framework beyond Southeast Asia to evaluate whether fragmentation typologies and path-dependent dynamics are consistent across different socio-ecological and governance contexts. Comparative studies in regions such as West Africa, Latin America, and small island systems would help determine whether dominant fragmentation pathways reflect region-specific pressures or broader patterns of urban coastal change. Such comparisons would strengthen the external validity of the framework and clarify how local drivers shape fragmentation trajectories.

Further work can integrate fragmentation trajectories with explicit land-use, infrastructure, and policy data to better understand the mechanisms underlying observed spatiotemporal transformations. Linking trajectory patterns with data on urban expansion, aquaculture development, transportation networks, and coastal engineering would help identify how specific disturbances and early fragmentation states influence long-term patch outcomes. This integration would improve the ability to connect observed change signatures to their underlying drivers.

Expanding temporal coverage through earlier baselines or higher-frequency observations would also

improve detection of short-term dynamics and transitions that are not captured in coarser time intervals. Finer temporal resolution would allow more detailed reconstruction of fragmentation pathways and better identification of transitional states.

Finally, applying this spatiotemporal and information-theoretic framework to other fragmented ecosystems would help evaluate its broader applicability. Systems such as urban wetlands, riparian forests, and coastal marshes provide opportunities to test whether trajectory-based fragmentation patterns and historical dependence are consistently observed across human-modified landscapes. Such cross-system applications would help clarify how landscape structure, disturbance regimes, and management contexts influence the emergence and strength of path-dependent dynamics.

Conclusions

This study, based on urban mangrove systems in Southeast Asia, frames fragmentation as a historically structured and process-based phenomenon. Fragmentation unfolds through sequences of spatiotemporal transformations that shape how landscape patterns develop and persist under cumulative disturbance. Tracing these sequences provides a more detailed understanding of landscape change than single-time spatial summaries alone.

Using lineage-based reconstruction of patch trajectories, this study shows that earlier fragmentation states reduce uncertainty in later patch outcomes beyond the immediately preceding state within baseline-traceable trajectories. These results provide evidence of weak-to-moderate path dependence along baseline-traceable patch trajectories. The strength of this dependence varies across urban contexts and reflects differences in landscape structure, disturbance regimes, and development histories.

Path dependence is reflected in the organization of fragmentation pathways, influencing how sequences of spatiotemporal transformations unfold over time. In complex urban coastal systems shaped by interacting infrastructural, environmental, and socio-political drivers, fragmentation dynamics remain contingent and responsive to ongoing disturbances and interventions.

The identification of recurring fragmentation pathways and their associations with patch persistence and loss highlights the importance of incorporating temporal structure into fragmentation analysis. Understanding fragmentation requires attention to both current spatial patterns and the sequences of transformations that produce them.

Overall, this study demonstrates how integrating trajectory-based analysis with information-theoretic approaches can support more nuanced interpretation of landscape change and provide a stronger foundation for studying fragmentation dynamics in human-modified ecosystems.

Author contributions A.G.C.G. conceived the study, developed the methodology, curated and analyzed the data, implemented the software and validation procedures, produced all visualizations, and wrote the original manuscript draft. K.C.S. provided supervision and conceptual guidance throughout the research process and contributed to manuscript review and editing.

Funding This research was supported by the Three Cairns Climate Program for the Global South, the Three Cairns Summer Fellowship, and the Jubitz Family Endowment for Research Internships Fund. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Data availability The data and code supporting the findings of this study are publicly available at: <https://github.com/greekmyth2298/Mangrove-Fragmentation-Typologies>. This repository provides all processed datasets and scripts required to reproduce the analyses presented in this study. The repository includes the datasets used for model development and evaluation, including the training dataset for typology classification, the external validation dataset, and the reconstructed trajectory dataset used for path-dependence analysis. The reconstructed trajectory dataset was generated through GIS-based patch tracking and manual validation prior to the analytical workflow. All scripts in the repository enable full reproduction of the analytical pipeline, from data processing to final model outputs. Mangrove extent data used in this study are publicly available through the Global Mangrove Watch v3.0 dataset at <https://zenodo.org/records/6894273>. Urban area boundaries were derived from the Morphological Urban Areas dataset, available at <https://data.mendeley.com/datasets/v3p8gk5724/1>.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as

long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Adler K, Jedicke E (2022) Landscape metrics as indicators of avian community structures – a state of the art review. *Ecol Indic* 145:109575
- Best ÜSN, van der Wegen M, Dijkstra J, Reyns J, van Prooijen BC, Roelvink D (2022) Wave attenuation potential, sediment properties and mangrove growth dynamics data over Guyana's intertidal mudflats: assessing the potential of mangrove restoration works. *Earth Syst Sci Data* 14(5):2445–2462
- Bimrah K, Dasgupta R, Hashimoto S, Saizen I, Dhyani S (2022) Ecosystem services of mangroves: a systematic review and synthesis of contemporary scientific literature. *Sustainability* 14(19):12051
- Blanco-Libreros JF, Ramírez-Ruiz K (2021) Threatened mangroves in the Anthropocene: habitat fragmentation in urban coastalscapes of *Pelliciera* spp. (Tetrameristaceae) in northern South America. *Front Mar Sci* 8:670354
- Branoff BL (2020) The role of urbanization in the flooding and surface water chemistry of Puerto Rico's mangroves. *Hydrol Sci J* 65(8):1326–1343
- Bryan-Brown DN, Connolly RM, Richards DR, Adame F, Friess DA, Brown CJ (2020) Global trends in mangrove forest fragmentation. *Sci Rep* 10(1):7117
- Bunting P, Rosenqvist A, Hilarides L, Lucas RM, Thomas N, Tadono T, Worthington TA, Spalding M, Murray NJ, Rebelo L-M (2022) Global mangrove extent change 1996–2020: Global Mangrove Watch version 3.0. *Remote Sens* 14(15):3657
- Carlucci M, Salvati L (2023) Assessing path-dependent urban growth with geographically weighted regressions. *Environ Impact Assess Rev* 98:106920
- Chase JM, Knight TM (2013) Scale-dependent effect sizes of ecological drivers on biodiversity: why standardised sampling is not enough. *Ecol Lett* 16(s1):17–26
- Cushman SA, McGarigal K, Neel MC (2008) Parsimony in landscape metrics: strength, universality, and consistency. *Ecol Indic* 8(5):691–703
- Dormann CF, McPherson JM, Araújo MB, Bivand R, Bolliger J, Carl G, Davies RG, Hirzel A, Jetz W, Kissling WD, Kühn I, Ohlemüller R, Peres-Neto PR, Reineking B, Schröder B, Schurr FM, Wilson R (2007) Methods to account for spatial autocorrelation in the analysis of species distributional data: a review. *Ecography* 30(5):609–628
- Erftemeijer PLA, Price BA, Ito S, Yamamoto H, Agastian T, Cambridge ML (2021) Salvaging and replanting 300 mangrove trees and saplings in the arid Arabian Gulf. *Mar Freshw Res* 72(11):1577–1587
- Fahrig L (2013) Rethinking patch size and isolation effects: the habitat amount hypothesis. *J Biogeogr* 40(9):1649–1663
- Fardila D, Kelly LT, Moore JL, McCarthy MA (2017) A systematic review reveals changes in where and how we have studied habitat loss and fragmentation over 20 years. *Biol Conserv* 212:130–138
- Goldberg L, Lagomasino D, Thomas N, Fatoyinbo T (2020) Global declines in human-driven mangrove loss. *Glob Change Biol* 26(10):5844–5855
- Haddad NM, Brudvig LA, Clobert J, Davies KF, Gonzalez A, Holt RD, Lovejoy TE, Sexton JO, Austin MP, Collins CD, Cook WM, Damschen EI, Ewers RM, Foster BL, Jenkins CD, King AJ, Laurance WF, Levey DJ, Margules CR, Melbourne BA, Nicholls AO, Orrock JL, Song D, Townshend JR (2015) Habitat fragmentation and its lasting impact on Earth's ecosystems. *Sci Adv* 1(2):e1500052
- Hadley AS, Betts MG (2016) Refocusing habitat fragmentation research using lessons from the last decade. *Curr Landsc Ecol Rep* 1(2):55–66
- Harris RJ, Milbrandt EC, Everham EM, Bovard BD (2010) The effects of reduced tidal flushing on mangrove structure and function across a disturbance gradient. *Estuaries Coasts* 33(5):1176–1185
- Henderson CJ, Gilby BL, Schlacher TA, Connolly RM, Sheaves M, Flint N, Borland HP, Olds AD (2019) Contrasting effects of mangroves and armoured shorelines on fish assemblages in tropical estuarine seascapes. *ICES J Mar Sci* 76(4):1052–1061
- Johnstone JF, Allen CD, Franklin JF, Frelich LE, Harvey BJ, Higuera PE, Mack MC, Meentemeyer RK, Metz MR, Perry GL, Schoennagel T, Turner MG (2016) Changing disturbance regimes, ecological memory, and forest resilience. *Front Ecol Environ* 14(7):369–378
- Magnan AK, Oppenheimer M, Garschagen M, Buchanan MK, Duvat VKE, Forbes DL, Ford JD, Lambert E, Petzold J, Renaud FG, Sebesvari Z, Van De Wal RSW, Hinkel J, Pörtner H-O (2022) Sea level rise risks and societal adaptation benefits in low-lying coastal areas. *Sci Rep* 12(1):10677
- Markham K, Frazier AE, Singh KK, Madden M (2023) A review of methods for scaling remotely sensed data for spatial pattern analysis. *Landsc Ecol* 38(3):619–635
- McGarigal K, Cushman SA (2002) Comparative evaluation of experimental approaches to the study of habitat fragmentation effects. *Ecol Appl* 12(2):335–345
- McGarigal K, Wan HY, Zeller KA, Timm BC, Cushman SA (2016) Multi-scale habitat selection modeling: a review and outlook. *Landsc Ecol* 31(6):1161–1175
- McGarigal K, Marks BJ (1995) FRAGSTATS: Spatial pattern analysis program for quantifying landscape structure (Gen. Tech. Rep. PNW-GTR-351). U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station. <https://doi.org/10.2737/PNW-GTR-351>
- McGarigal K, Cushman S, Regan C (2005) Quantifying terrestrial habitat loss and fragmentation: A protocol. University

- of Massachusetts, Department of Natural Resources Conservation. https://www.fs.usda.gov/rm/pubs_journals/2005/rmrs_2005_mcgarrigal_k001.pdf
- Murphy SJ (2021) Sampling units derived from geopolitical boundaries bias biodiversity analyses. *Glob Ecol Biogeogr* 30(9):1876–1888
- Nowosad J, Stepinski TF (2019) Information theory as a consistent framework for quantification and classification of landscape patterns. *Landsc Ecol* 34(9):2091–2101
- Peters DPC, Lugo AE, Chapin FS, Pickett STA, Duniway M, Rocha AV, Swanson FJ, Laney C, Jones J (2011) Cross-system comparisons elucidate disturbance complexities and generalities. *Ecosphere* 2(7):art81
- Rani V, Nandan SB, Jayachandran PR, Preethy CM, Sreelekshmi S, Joseph P, Asha CV (2023) Carbon stock in biomass pool of fragmented mangrove habitats of Kochi, Southern India. *Environ Sci Pollut Res Int* 30(43):96746–96762
- Reyes GJ, Smyth AR, Qiu J, Reynolds LK (2024) Highly urbanized mangrove areas are small in size, fragmented, and missed by large-scale mapping efforts. *Wetlands* 44(7):103
- Richards DR, Friess DA (2016) Rates and drivers of mangrove deforestation in Southeast Asia, 2000–2012. *Proc Natl Acad Sci U S A* 113(2):344–349
- Roy Chowdhury R, Turner BL (2019) The parallel trajectories and increasing integration of landscape ecology and land system science. *J Land Use Sci* 14(2):135–154
- Rutledge D (2003) Landscape indices as measures of the effects of fragmentation: Can pattern reflect process? DOC Science Internal Series 98. Department of Conservation. <https://www.doc.govt.nz/Documents/science-and-technical/DSIS98.pdf>
- Tappeiner U, Leitinger G, Zariņa A, Bürgi M (2021) How to consider history in landscape ecology: patterns, processes, and pathways. *Landsc Ecol* 36(8):2317–2328
- Taubenböck H, Weigand M, Esch T, Staab J, Wurm M, Mast J, Dech S (2019) A new ranking of the world's largest cities—do administrative units obscure morphological realities? *Remote Sens Environ* 232:111353
- Turner MG (2010) Disturbance and landscape dynamics in a changing world. *Ecology* 91(10):2833–2849
- Turner MG, Gardner RH (2015) *Landscape ecology in theory and practice: pattern and process*, 2nd ed. Springer. <https://doi.org/10.1007/978-1-4939-2794-4>
- Wan HY (2025) It's about time: temporal fragmentation and metrics for habitat change. *Landsc Ecol* 40(12):226
- Wang X, Blanchet FG, Koper N (2014) Measuring habitat fragmentation: an evaluation of landscape pattern metrics. *Methods Ecol Evol* 5(7):634–646
- Wu Z, Yu H, Shola AA, Chang X, Jiang W (2024) Numerical investigation of land reclamation effects on hydrodynamics and mangroves in Shacheng Bay for the last 36 years. *J Ocean Univ China* 23(6):1436–1448
- Zariņa A (2013) Path dependence and landscape: initial conditions, contingency and sequences of events in Latgale, Latvia. *Geogr Ann B Hum Geogr* 95(4):355–373
- Zhang Y, Wang P, Wang T, Li J, Li Z, Teng M, Gao Y (2020) Using vegetation indices to characterize vegetation cover change in the urban areas of southern China. *Sustainability* 12(22):9403
- Zhang Z, He G, Cai W, Zhu Q, Liu X, Ding F, Cai Y (2024) Spatial planning constraints will mitigate the fragmentation trajectory of natural and semi-natural landscapes: a case of Lushan City, China. *Landsc Ecol* 39(3):59
- Zuur AF, Ieno EN, Walker N, Saveliev AA, Smith GM (2009) *Mixed effects models and extensions in ecology with R*. Springer, New York. <https://doi.org/10.1007/978-0-387-87458-6>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.